

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
K. K. BIRLA GOA CAMPUS

I/II Semester 20 -20

MIDTERM TEST / COMPREHENSIVE EXAMINATION

Sec. No. _____

ID No. 2023 A7PSO 466G Course No. 20 CS F351
Name Arnav Adivi Course Title Theory of Computation
No. of Supplementary copies attached: _____ Date/Day 9/10/25

Question No.	Marks	Student's request for rechecking with remarks	Examiner's remarks
1.	20		
2.	8		
3.	10		
4.	12		
5.	10		
6.			
7.			
8.			
Total			
(60) (in figures)		(in words)	

INSTRUCTIONS TO CANDIDATES

Examiner's Signature

(1) Write clearly and legibly. (2) Enter all the required details on the cover of every answer book. (3) The question number given in the answersheet by the student while answering the question should be the same as in the question paper. (4) Write on both sides of the sheet in the answer book. Rough work if any, should be done at the bottom of the page. Finally cross it out and draw a horizontal line to separate it from the rest of the material on the page. (5) Any answer crossed out by the student will not be examined by the examiner. (6) A supplementary answer book should not be asked for until the first answer book is filled up. (7) No sheet should be torn from the answer book. (8) Use of any unfair means will make the candidate liable to disciplinary actions. (9) No paper should be brought in the examination hall for scribbling on. (10) A student should not leave the examination hall without handing over the answer book to invigilator on duty.

Theory of Computation

3. Assume L is regular. Then there exists a $n \geq 1$ for which the statement of the lemma holds.

The statement applies to the string $0^n 1^n 0^{2n}$ since $2n \geq n+n$ and $|0^n 1^n 0^{2n}| = 4n \geq n$.

Hence $0^n 1^n 0^{2n} = xyz$ for some x, y, z where $|xy| \leq n$, $y \neq \epsilon$, and $xy^i z \in L$ for each $i \geq 0$. Since $|xy| \leq n$,

xy consists of all zeroes, and so $y = 0^r$

for some $r \geq 1$. Hence $0^n 1^n 0^{2n} = 0^{n-r} 0^r (1^n 0^{2n})$

for and in this case $0^{n-r} (0^r)^5 (1^n 0^{2n}) \in L$.

However, ~~$0^{n-r} (0^r)^5 1^n 0^{2n} \notin L$~~

$0^{n-r} (0^r)^5 1^n 0^{2n} = 0^{n+4r} 1^n 0^{2n}$. If this is in

L , $2n \geq 2n + 4r \Rightarrow 4r \leq 0 \Rightarrow r \leq 0$. This is not possible as $r \geq 1$. Hence L is not regular.

4.

$$a) (b \cup ab \cup aab)^* aaa (b \cup ba \cup baa)^*$$

b)

$x \approx_L y$ ~~iff~~ by definition when

$$xz \in L \text{ iff } yz \in L \text{ for all } z \in \Sigma^*.$$

In the case $z=e$, we get that
any two equivalent strings for this language
either both end in a or both do not
end in a .

First equivalence class:

$$[a] = \cancel{\{a\}} \cup \cancel{\{a, b\}^* a} = \{a, b\}^* a \quad \checkmark$$

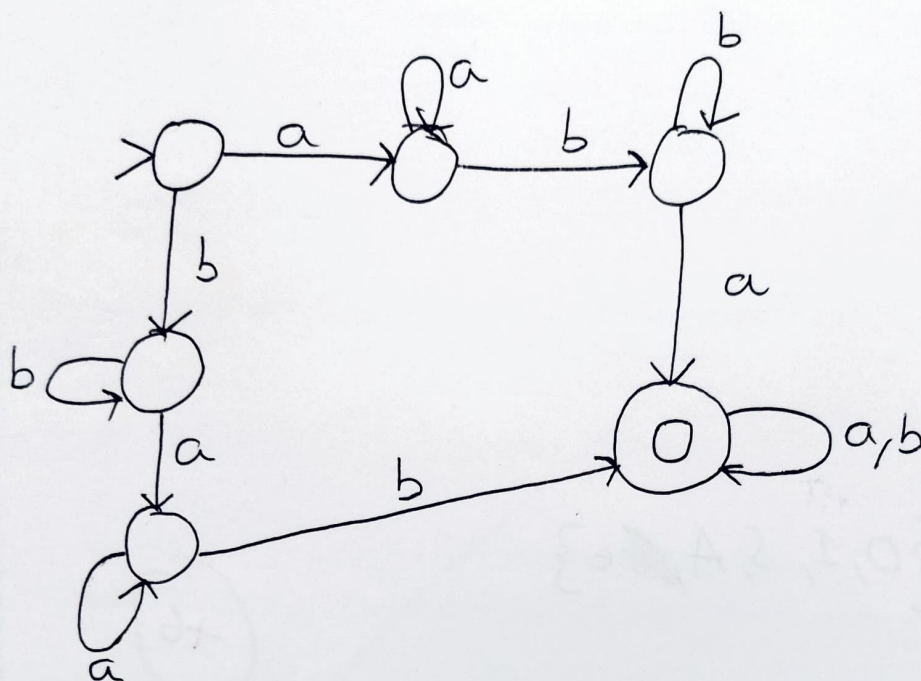
~~Second~~

Second equivalence class:

$$[e] = \{e\} \cup (\{a, b\}^* a \{b\}) = \{a, b\}^* a \{b, e\} \quad \checkmark$$

2.

8



5.

a)

~~We shall show that $L(G) = L$. Suppose $w \in L(G)$.~~

~~None of the rules change the numbers of zeroes and ones~~

~~+3~~ +4

No it does not. Consider $1100 \in L$. Then

$n_0(1100) = n_1(1100) = 2$, but 1100 cannot be generated. ~~It~~

add more justification
in context with the grammar

#)

b)

$$V := \{0, 1, S, A, \cancel{B}, e\}$$

+6

$$\Sigma := \{0, 1, e\}$$

R:

$$S \rightarrow \cancel{AB} AO$$

$$S \rightarrow \cancel{AB} OA$$

~~$$S \rightarrow 01B$$~~
~~$$S \rightarrow 10B$$~~
~~$$S \rightarrow AB1$$~~

$$A \rightarrow 1AO$$

$$A \rightarrow 1OA$$

$$A \rightarrow A1O$$

$$A \rightarrow OA1$$

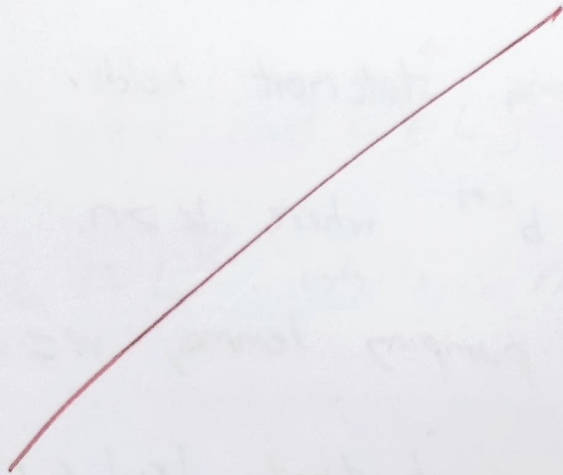
$$A \rightarrow O1A$$

$$A \rightarrow A01$$

$$A \rightarrow e$$



$$S=S$$



1

a) False. Let $L_1 := \{a^n \in \{a\}^* : n \text{ is prime}\}$

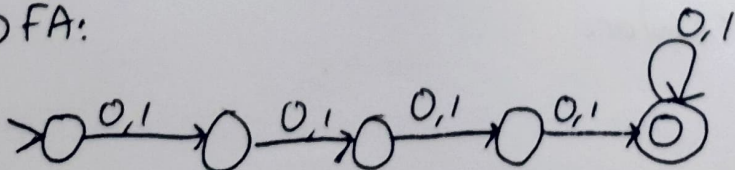
and $L_2 := \{a^n \in \{a\}^* : n \text{ is not prime}\}$

$L_1 \cup L_2 = \{a\}^*$ which is regular but

neither L_1 nor L_2 are regular.

~~Verification: $L_2 = \{a^n \in \{a\}^* : n \text{ is not prime}\}$~~

b) Yes, because it ^{is} ~~can be~~ ^{accepted} ~~represented~~ by the
 $\hookrightarrow$ DFA:



⊗ C) Not regular. Suppose L is regular. Then

5
there exists a $n \geq 1$ for which the pumping lemma statement holds. Consider the

string $w = a^k b^{k+1}$ where $k > n$. Clearly $w \in L$.

By the pumping lemma, $w = xyz$ for some $x, y \neq \epsilon, z$ such that $|xy| \leq n$ and $xy^i z \in L$

for all $i \geq 0$. Since $k > n$, xy consists of

all a 's. Then $w = \underbrace{a^m}_x \underbrace{a^r}_y \underbrace{a^{k-m-r}}_z b^{k+1}$

for some m, r where $r > 1$. Then

$$\begin{aligned} xy^{k+1}z &= a^m a^{(k+1)r} a^{k-m-r} b^{k+1} \\ &= a^{k+kr} b^{k+1} = a^{r(k+1)} b^{k+1} \in L \end{aligned}$$

⊗ Hence $\gcd(r(k+1), k+1) = 1$. But this is not possible as clearly $\gcd(r(k+1), k+1) \geq k+1 > n+1 > 1$.
 L is not regular.

d) Yes, as it is the intersection of the regular
languages L and L^R

$$x \in \{w : w \in L \text{ and } w^R \in L\} \text{ iff } (x \in L \text{ and } x \in L^R)$$

i.e. $x \in L \cap L^R$ which is regular